



Can we automate tuning an EKF?

- “Tuning” a KF is the process of selecting $\Sigma_{\tilde{x},0}^+$, $\Sigma_{\tilde{w}}$, and $\Sigma_{\tilde{v}}$.
- For linear systems, perhaps we can find $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ from sensor datasheets, or from independent measurements.
- But, $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ also describe, to an extent, inaccuracies in the state and measurement equations, and so are not trivial to determine even in the linear case.
- For nonlinear systems, the overall impact of w_k and v_k can be additionally difficult to quantify due to the nonlinearities of the state and measurement equations.
- It would be very nice if we could automate a way to estimate $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ from data.



Literature search

- The earliest work that I’m aware of on this topic is a paper by Mehra.¹ This paper is still heavily cited.
 - The derivation is surprisingly complicated.
- Different variations of Mehra’s approach appear in the literature, because of making different assumptions in different steps.
- Two of the best references I have found for putting the best practices of all of these into a single document are by Fraser.²

¹Mehra, Raman. “On the identification of variances and adaptive Kalman filtering.” *IEEE Transactions on automatic control* 15(2), 1970, 175–184.

²Cory T. Fraser and Steve Ulrich. “Adaptive extended Kalman filtering strategies for spacecraft formation relative navigation.” *Acta Astronautica*, 178, 2021, 700–721.

Cory T. Fraser, *Adaptive Extended Kalman Filtering Strategies for Autonomous Relative Navigation of Formation Flying Spacecraft*, MSAE Thesis, Carleton University, 2018.



Basic idea

- I’m not going to present the full derivation here. If you are interested, please confer documents cited on the prior slide.
 - But, I would like to give a “sketch” of the idea.
- We treat $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ as unknown parameters of the system model. We can define this set of parameters as $\Theta = \{\Sigma_{\tilde{w}}, \Sigma_{\tilde{v}}\}$.
- We wish to find the $\Sigma_{\tilde{w}}, \Sigma_{\tilde{v}}$ that maximize the likelihood of Θ given the set of measurements $\mathbb{Z}_k$:

$$\Theta^* = \arg \max_{\Theta} \{L(\Theta | \mathbb{Z}_k)\}.$$

- To do this, we assume that Θ has a Gaussian distribution and substitute the Gaussian pdf in the computation of likelihood, and proceed from there.



Not a proof: Motivating form of $\Sigma_{\tilde{w}}$

- If we assume w_k to be additive, then the state equation says:

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}.$$

- Using EKF as a basis, consider Steps 1a and 2b:

$$\begin{aligned}\hat{x}_k^- &= \mathbb{E}[x_k | \mathbb{Z}_{k-1}] \approx f(\hat{x}_{k-1}^+, u_{k-1}) + \bar{w}_{k-1} \\ \hat{x}_k^+ &= \mathbb{E}[x_k | \mathbb{Z}_k] \approx \hat{x}_k^- + L_k \underbrace{(z_k - \hat{z}_k)}_{\mu_k}.\end{aligned}$$

- If we assume $\hat{x}_k^+ \approx x_k$, then $\hat{x}_k^+ - \hat{x}_k^- \approx w_k - \bar{w}_k = \tilde{w}_{k-1}$.
 - But, we also see that $\hat{x}_k^+ - \hat{x}_k^- = L_k \mu_k$ (where μ_k is the innovation).
 - So, we can approximate: $\tilde{w}_{k-1} \approx \hat{x}_k^+ - \hat{x}_k^- = L_k \mu_k$.
- The covariance that we seek:

$$\Sigma_{\tilde{w}} = \mathbb{E}[\tilde{w}_{k-1} \tilde{w}_{k-1}^T] \approx \mathbb{E}[L_k \mu_k \mu_k^T L_k^T].$$



Not a proof: Motivating form of $\Sigma_{\tilde{v}}$

- Again using EKF as a basis, consider Step 2a:

$$L_k = \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1},$$

where $\Sigma_{\tilde{z}} = \hat{C}_k \Sigma_{\tilde{x},k}^- \hat{C}_k^T + \Sigma_{\tilde{v}}$ is the covariance of the innovation μ_k .

- So, we have that $\Sigma_{\tilde{v}} = \mathbb{E}[\mu_k \mu_k^T] - \hat{C}_k \Sigma_{\tilde{x},k}^- \hat{C}_k^T$.
 - Some adaptive EKFs use this relationship.
 - However, the subtraction can lead to non-positive-definite $\Sigma_{\tilde{v}}$ in practice.
- A more robust solution defines the residual: $r_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
 - Note that the residual uses $\hat{x}_k^+$ and the innovation uses $\hat{x}_k^-$.
 - Then, it is possible to derive $\Sigma_{\tilde{v}} = \mathbb{E}[r_k r_k^T + \hat{C}_k \Sigma_{\tilde{x},k}^+ \hat{C}_k^T]$.



Smoothing the estimates of $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$

- Instantaneous estimates of $\Sigma_{\tilde{w},k}$ and $\Sigma_{\tilde{v},k}$ using these equations tend to be very noisy and introduce instability into the adaptive EKF (AEKF) solution.
- Authors use ad-hoc methods to smooth estimates of $\Sigma_{\tilde{w},k}$ and $\Sigma_{\tilde{v},k}$ either by averaging over a buffer of $\Sigma_{(\cdot),k}$ values or by using a simple one-pole filter:

$$\Sigma_{(\cdot),k} = \alpha \Sigma_{(\cdot),k-1} + (1 - \alpha) [\text{New value of } (\cdot)_k].$$

- I find that smoothing is very important to getting AEKF to work.
- My experience is that the method to estimate $\Sigma_{\tilde{v}}$ tends to work quite well.
- But I often have trouble with the method to estimate $\Sigma_{\tilde{w}}$.
 - I think that this is because $\hat{x}_k^+ - \hat{x}_k^-$ is not exactly equal to $\tilde{w}_{k-1}$.
- The next lesson will introduce AEKF code and share an example: You will see how I modified one step to make it work well for that example.



Summary

- The AEKF seeks to make estimates $\Sigma_{\tilde{w},k}$ and $\Sigma_{\tilde{v},k}$ of the process-noise and sensor-noise covariance matrices.
- My own experiences with the AEKF have been mixed: sometimes I get good results and other times the filter becomes unstable.
- But, AEKF is very popular in the literature and it is a compelling approach to finding $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ if it works well for your nonlinear system.
- It is not difficult to implement, as you will learn in the next lesson, and so it is worth trying.